

A Conditional Solution to Iima–Yoshino Problem 2.3

An infinite Gröbner degeneration for the first Rogers–Ramanujan identity

Working research note

2 September 2026

Abstract

We give an explicit solution to Problem 2.3 of Iima and Yoshino over every field k of characteristic different from 5 that contains an element c satisfying $c^2 + c = 1$. For the weighted polynomial ring $S = k[x_1, x_2, \dots]$, $\deg x_i = i$, we construct a homogeneous ideal I such that

$$S/I \cong k[x_i : i \equiv \pm 1 \pmod{5}], \quad \text{in}(I) = (x_i^2, x_i x_{i+1} : i \geq 1).$$

The construction is governed by the Lyness five-cycle. A single explicit periodic syzygy both proves that the quotient is polynomial and supplies standard representations for all non-coprime S -pairs. Thus the proof is purely algebraic and does not use the Rogers–Ramanujan identity; conversely, comparison of the quotient with its initial algebra recovers that identity. In each weighted degree, the invertible normal-form matrix has a perfect matching in its support, yielding a bijection between the two partition sets (and an effective procedure over an effectively presented coefficient field). Unlike the binomial examples of Iima and Yoshino, however, an individual normal form need not be a single monomial. The construction applies over $\mathbb{C}$ and $\mathbb{Q}(\sqrt{5})$, but it does not presently solve the arbitrary-field version, in particular over $\mathbb{Q}$ or in characteristic 5.

1 The problem and the result

Let

$$S = k[x_1, x_2, \dots], \quad \deg x_i = i.$$

Iima and Yoshino asked for an ideal $I \subset S$ and a monomial order such that

$$S/I \cong k[x_i : i \equiv \pm 1 \pmod{5}], \quad \text{in}(I) = J := (x_i^2, x_i x_{i+1} : i \geq 1). \quad (1)$$

The standard monomials modulo J encode partitions whose adjacent parts differ by at least two, while monomials in the polynomial ring on the right-hand side encode partitions into parts congruent to ± 1 modulo 5.

Our construction requires the following coefficient-field hypothesis.

Standing hypothesis. The field k has characteristic different from 5, and $c \in k$ satisfies

$$c^2 + c = 1. \quad (2)$$

In particular $c(c+1) = 1$, so c is a unit. In characteristic zero this hypothesis holds over $\mathbb{Q}(\sqrt{5})$, with $c = (-1 \pm \sqrt{5})/2$.

Define two sequences indexed by $\mathbb{Z}/5\mathbb{Z}$:

$$\begin{array}{c|ccccc} j & 0 & 1 & 2 & 3 & 4 \\ \hline s_j & 2 & c & -1-c & -1-c & c \\ \eta_j & 0 & 1 & c & -c & -1. \end{array} \quad (3)$$

All subscripts on s and η below are read modulo 5.

For $n \geq 2$, put

$$g_n = (s_n - c)x_n + \sum_{1 \leq i < n/2} s_{n-2i}x_i x_{n-i} + \begin{cases} x_{n/2}^2, & 2 \mid n, \\ 0, & 2 \nmid n. \end{cases} \quad (4)$$

Thus each g_n is homogeneous of weighted degree n . Let

$$D = \{n \geq 2 : n \equiv 0, 2, 3 \pmod{5}\}, \quad I = (g_n : n \in D). \quad (5)$$

The first relations are

$$\begin{aligned} g_2 &= x_1^2 - (1 + 2c)x_2, \\ g_3 &= cx_1x_2 - (1 + 2c)x_3, \\ g_4 &= x_2^2 - (1 + c)x_1x_3, \\ g_5 &= cx_2x_3 - (1 + c)x_1x_4 + (2 - c)x_5, \\ g_6 &= x_3^2 - (1 + c)x_2x_4 + cx_1x_5. \end{aligned}$$

Theorem 1.1. *Under the standing hypothesis, the ideal I in (5) satisfies (1). More precisely, for the monomial order defined in Section 4, the family*

$$\mathcal{G} = \{g_{2m} : m \geq 1\} \cup \{c^{-1}g_{2m+1} : m \geq 1\}$$

is the reduced Gröbner basis of I , and

$$\text{lm}(g_{2m}) = x_m^2, \quad \text{lm}(g_{2m+1}) = x_m x_{m+1}.$$

Furthermore, the residue classes of the variables x_n with $n \equiv 1, 4 \pmod{5}$ freely generate S/I as a polynomial algebra.

2 The periodic syzygy

The core of the proof is the following identity.

Lemma 2.1 (five-periodic syzygy). *For every $N \geq 2$,*

$$-(\eta_N + c\eta_{2N})g_N + \sum_{r=1}^{N-2} \eta_{4N-2r} x_r g_{N-r} = 0. \quad (6)$$

The coefficient of g_N is

$$-(\eta_N + c\eta_{2N}) = \begin{cases} 0, & N \equiv 0, 2, 3 \pmod{5}, \\ c - 2, & N \equiv 1 \pmod{5}, \\ 2 - c, & N \equiv 4 \pmod{5}. \end{cases} \quad (7)$$

Moreover,

$$\eta_{4N-2r} = 0 \iff r \equiv 2N \pmod{5}. \quad (8)$$

Proof. The values in (7) and (8) follow immediately from the table (3) and $c^2 + c = 1$. We give a conceptual derivation of (6); it can alternatively be verified directly by the same five-residue calculation.

Pass temporarily to a field extension containing a root ζ of $z^2 - cz + 1$. Since $\zeta + \zeta^{-1} = c$, equation (2) gives

$$\zeta^2 + \zeta + 1 + \zeta^{-1} + \zeta^{-2} = 0.$$

Thus $\zeta^5 = 1$; and $\zeta \neq 1$ because otherwise $c = 2$ would force $5 = 0$. Hence ζ is a primitive fifth root of unity. The first row of (3) is therefore

$$s_j = \zeta^j + \zeta^{-j}.$$

Set

$$F(t) = 1 + \sum_{n \geq 1} x_n t^n$$

and

$$R(t) = F(\zeta^{-1}t)F(\zeta t) - (1 - c + cF(t)).$$

Pairing the terms indexed by i and $n - i$ shows directly that

$$R(t) = \sum_{n \geq 2} g_n t^n. \tag{9}$$

Let $\gamma = c^{-1} = 1 + c$, $G = F/c$, and

$$E(t) = G(\zeta^{-1}t)G(\zeta t) - 1 - G(t) = c^{-2}R(t).$$

For indices modulo 5, write

$$G_j(t) = G(\zeta^j t), \quad E_j(t) = E(\zeta^j t).$$

Then

$$E_j = G_{j-1}G_{j+1} - 1 - G_j.$$

The following identity is formal: expanding its four summands makes every term cancel,

$$G_2E_4 - G_3E_1 - E_2 + E_3 = 0. \tag{10}$$

Write $E(t) = \sum_{n \geq 2} e_n t^n$, where $e_n = c^{-2}g_n$. The coefficient of t^N in (10) is

$$A_N e_N + \frac{1}{c} \sum_{r=1}^{N-2} C_{N,r} x_r e_{N-r} = 0, \tag{11}$$

where

$$\begin{aligned} A_N &= \gamma(\zeta^{4N} - \zeta^N) - \zeta^{2N} + \zeta^{3N}, \\ C_{N,r} &= \zeta^{4N-2r} - \zeta^{N+2r}. \end{aligned}$$

Put $\Delta = \zeta - \zeta^{-1}$. The second row in (3) satisfies

$$\eta_j = \frac{\zeta^j - \zeta^{-j}}{\Delta}.$$

Since $(4N - 2r) + (N + 2r) = 5N$, we have

$$C_{N,r} = \Delta \eta_{4N-2r}.$$

Similarly,

$$A_N = -\Delta(\gamma\eta_N + \eta_{2N}).$$

Divide (11) by Δ , multiply by c , and use $c\gamma = 1$ and $e_n = c^{-2}g_n$. This gives exactly (6). Since the resulting identity has coefficients in k , it holds before the temporary field extension. $\square$

Remark 2.2 (Lyness interpretation). Modulo all the relations g_n , equation (9) says

$$G(\zeta^{-1}t)G(\zeta t) = 1 + G(t).$$

If $A_j = G(\zeta^j t)$, this is $A_{j-1}A_{j+1} = 1 + A_j$, the Lyness five-cycle. The birational map $(u, v) \mapsto (v, (1+v)/u)$ has order five. Identity (10) is a syzygetic form of this pentagon periodicity.

3 The polynomial quotient

We first verify that the ideal generated by the non-resonant relations has the required quotient.

Lemma 3.1. *The elements $1 + 2c$ and $2 - c$ are units in k .*

Proof. Using $c^2 = 1 - c$,

$$(1 + 2c)^2 = 5, \quad (2 - c)(3 + c) = 5.$$

Since $\text{char}(k) \neq 5$, the assertion follows. $\square$

Proposition 3.2. *There is a natural graded isomorphism*

$$S/I \cong k[x_n : n \equiv 1, 4 \pmod{5}].$$

Moreover, $I = (g_n : n \geq 2)$.

Proof. If $n \in D$, then (4) has the triangular form

$$g_n = a_n x_n + h_n(x_1, \dots, x_{n-1}),$$

where

$$a_n = \begin{cases} 2 - c, & n \equiv 0 \pmod{5}, \\ -1 - 2c, & n \equiv 2, 3 \pmod{5}. \end{cases}$$

Lemma 3.1 shows that a_n is a unit. We can therefore eliminate x_n recursively for all $n \in D$. More explicitly, map each x_n with $n \equiv 1, 4 \pmod{5}$ to an independent variable of the same degree, and define the image of a bad-residue variable recursively from $x_n = -a_n^{-1}h_n$. Conversely, eliminating the finitely many bad-residue variables occurring in any given polynomial produces a unique polynomial in the allowed variables. These mutually inverse procedures give the asserted isomorphism.

It remains to see that the omitted relations g_N , $N \equiv 1, 4 \pmod{5}$, already belong to I . By (7), their coefficient in (6) is $c - 2$ or $2 - c$, hence a unit. The remaining terms involve only g_{N-r} with $N - r < N$. Strong induction on N , starting with $g_1 = 0$, therefore gives $g_N \in I$. Hence $I = (g_n : n \geq 2)$. $\square$

For instance, the first two resonant relations are consequences of earlier relations through

$$\begin{aligned} (2 - c)g_4 - x_1g_3 + cx_2g_2 &= 0, \\ (c - 2)g_6 + cx_1g_5 - cx_3g_3 + x_4g_2 &= 0. \end{aligned}$$

4 The monomial order and leading terms

For a monomial $x^\alpha = \prod_i x_i^{\alpha_i}$, define

$$\text{wt}(\alpha) = \sum_i i\alpha_i, \quad \sigma(\alpha) = \sum_i i^2\alpha_i.$$

Fix any additive lexicographic order $>_{\text{tie}}$ on exponent vectors. Define $x^\alpha > x^\beta$ if, in order,

- (i) $\text{wt}(\alpha) > \text{wt}(\beta)$; or
- (ii) the weights agree and $\sigma(\alpha) < \sigma(\beta)$; or
- (iii) the weights and second moments agree and $x^\alpha >_{\text{tie}} x^\beta$.

Lemma 4.1. *This is a monomial order on S .*

Proof. All three comparison data are additive under multiplication, so the order is multiplicative. It is total after the final tie-break. Every nonempty set of monomials has a member of least weighted degree, and there are only finitely many monomials of a fixed weighted degree because $\deg x_i = i$. The restriction to that finite set has a least element. Hence the order is a well-order. $\square$

Lemma 4.2. *For every $m \geq 1$,*

$$\text{lm}(g_{2m}) = x_m^2, \quad \text{lm}(g_{2m+1}) = x_m x_{m+1}.$$

Their leading coefficients are respectively 1 and c .

Proof. Every term of g_n has weighted degree n . For a quadratic term $x_i x_{n-i}$ its second moment is

$$i^2 + (n-i)^2.$$

This strictly decreases as i approaches $n/2$. Thus the unique quadratic monomial of smallest second moment is x_m^2 for $n = 2m$, and $x_m x_{m+1}$ for $n = 2m + 1$. The linear term x_n has second moment n^2 and is lower in the order. Formula (4) gives coefficients 1 and $s_1 = c$ for the two balanced terms. $\square$

Every non-leading quadratic term in g_n has indices differing by at least two, and a linear term is never divisible by a quadratic monomial. Consequently, after dividing odd g_n by c , every tail term is standard with respect to J ; once the Gröbner property is proved, reducedness will follow immediately.

5 Buchberger's criterion

Put

$$q_n = \text{lm}(g_n) = x_{\lfloor n/2 \rfloor} x_{\lceil n/2 \rceil}.$$

The only pairs q_a, q_b that are not relatively prime are, up to symmetry,

$$(q_{2m}, q_{2m+1}), \quad (q_{2m+1}, q_{2m+2}), \quad (q_{2m-1}, q_{2m+1}), \quad (12)$$

where $m \geq 1$ in the first two families and $m \geq 2$ in the third.

For a term $x_r g_{N-r}$ in (6), the second moment of its product-leading monomial is

$$\rho_N(r) = r^2 + \left\lfloor \frac{N-r}{2} \right\rfloor^2 + \left\lceil \frac{N-r}{2} \right\rceil^2. \quad (13)$$

Lemma 5.1. *The relevant minima of (13) are as follows.*

- (i) *If $N = 3m + 1$, the minimum occurs exactly at $r = m, m + 1$.*
- (ii) *If $N = 3m + 2$, the minimum occurs exactly at $r = m, m + 1$.*
- (iii) *If $N = 3m$, the minimum occurs at $r = m$; the next smallest value occurs exactly at $r = m - 1, m + 1$.*

Proof. On each of the two parity classes of $N - r$, the function $\rho_N(r)$ is a strictly convex quadratic in r . Comparing the one or two integral points nearest its vertex gives the stated list. Equivalently, the claim follows by computing the successive differences $\rho_N(r+1) - \rho_N(r)$ separately for $N - r$ even and odd. $\square$

Proposition 5.2. *The family $\{g_n : n \geq 2\}$ is a Gröbner basis of I .*

Proof. We apply the infinite-variable Buchberger criterion of Iima and Yoshino. Pairs with relatively prime leading monomials satisfy the product criterion. It remains to treat the three families in (12).

For $N = 3m + 1$, Lemma 5.1 says that the two largest product terms in (6) occur at $r = m, m + 1$. Their coefficients from the η table are -1 and c , so the top part of the syzygy is

$$-x_m g_{2m+1} + c x_{m+1} g_{2m}.$$

The two leading terms are $-c x_m^2 x_{m+1}$ and $c x_m^2 x_{m+1}$; thus they cancel and form a nonzero scalar multiple of the S -polynomial of g_{2m} and g_{2m+1} .

For $N = 3m + 2$, the two largest product terms again occur at $r = m, m + 1$. Their top part is

$$-c x_m g_{2m+2} + x_{m+1} g_{2m+1},$$

whose leading terms cancel at the least common multiple $x_m x_{m+1}^2$.

Finally take $N = 3m$, $m \geq 2$. The unrestricted largest product would come from $r = m$ and would have leading monomial x_m^3 . It is absent, however, because

$$\eta_{4N-2m} = \eta_{10m} = 0.$$

By Lemma 5.1, the next and only maximal products occur at $r = m - 1, m + 1$. Their top part is

$$c x_{m-1} g_{2m+1} - c x_{m+1} g_{2m-1},$$

and the leading terms cancel at $x_{m-1} x_m x_{m+1}$.

In each case all other product terms in (6) are strictly below the corresponding least common multiple. The standalone term involving g_N is also lower: its leading monomial has second moment at least $N^2/2$, whereas the three least common multiples have second moments respectively

$$3m^2 + 2m + 1, \quad 3m^2 + 4m + 2, \quad 3m^2 + 2.$$

The resulting differences are positive for $m \geq 1$ in the first two cases and $m \geq 2$ in the third. Thus (6) is a standard representation of each non-coprime S -polynomial. Buchberger's criterion now proves the claim. $\square$

Proof of Theorem 1.1. Proposition 3.2 proves the quotient assertion and shows that the complete family $\{g_n : n \geq 2\}$ generates I . Lemma 4.2 and Proposition 5.2 give

$$\text{in}(I) = (x_m^2, x_m x_{m+1} : m \geq 1) = J.$$

After the odd relations are divided by c , the family is monic. As noted after Lemma 4.2, none of its tail monomials is divisible by a generator of J . Hence the displayed Gröbner basis is reduced. $\square$

6 Consequences and limitations

The Rogers–Ramanujan identity

The monomials outside J are precisely the monomials $x_{\lambda_1} \cdots x_{\lambda_r}$ with $\lambda_i - \lambda_{i+1} \geq 2$. Since passing to an initial ideal preserves each finite-dimensional weighted component, Theorem 1.1 gives

$$\prod_{j \equiv \pm 1 (5)} \frac{1}{1 - t^j} = H_{S/I}(t) = H_{S/J}(t),$$

which is the partition form of the first Rogers–Ramanujan identity. Thus the identity is a consequence, rather than an input, of the construction.

The field hypothesis

The original paper works with an arbitrary field. Our construction needs a root of $c^2 + c - 1$ and the invertibility of 5. It therefore gives a complete solution over $\mathbb{C}$, $\mathbb{Q}(\sqrt{5})$, and more generally under the standing hypothesis, but not yet over $\mathbb{Q}$ or every preassigned field. In characteristic 5, the coefficients needed for triangular elimination vanish. This shows that the present construction degenerates; it does not prove that no different construction can exist.

An algorithmic bijection, but not monomial normal forms

Let $P(n)$ denote the partitions of n into parts congruent to ± 1 modulo 5, and let $Q(n)$ denote the partitions $\mu = (\mu_1, \dots, \mu_r)$ satisfying $\mu_i - \mu_{i+1} \geq 2$. Theorem 1.1 does give a genuine one-to-one correspondence between these sets, although one additional finite linear-algebra step may be needed after division.

Proposition 6.1. *For every n , there is a bijection*

$$\pi_n : P(n) \longrightarrow Q(n)$$

such that $x^{\pi_n(\lambda)}$ occurs with nonzero coefficient in the normal form of x^λ for every $\lambda \in P(n)$. The bijection can be obtained from Gröbner division followed by choosing a perfect matching in a finite bipartite graph. If arithmetic and equality in k are effective, this is an algorithm.

Proof. By the polynomial-quotient assertion, the residue classes

$$\mathcal{B}_P(n) = \{x^\lambda : \lambda \in P(n)\}$$

form a basis of $(S/I)_n$. By the initial-ideal assertion, the standard monomials

$$\mathcal{B}_Q(n) = \{x^\mu : \mu \in Q(n)\}$$

form a basis of $(S/J)_n$. The normal-form map is an isomorphism between these finite-dimensional spaces. Write

$$\text{NF}(x^\lambda) = \sum_{\mu \in Q(n)} a_{\lambda, \mu} x^\mu, \quad A_n = (a_{\lambda, \mu})_{\lambda \in P(n), \mu \in Q(n)}.$$

The matrix A_n is square and invertible. Hence in the determinant expansion of A_n there is at least one permutation π_n for which

$$\prod_{\lambda \in P(n)} a_{\lambda, \pi_n(\lambda)} \neq 0.$$

Equivalently, form the bipartite graph with vertex classes $P(n)$ and $Q(n)$ and an edge $\lambda \sim \mu$ exactly when $a_{\lambda, \mu} \neq 0$. The preceding nonzero determinant term is a perfect matching. This proves existence. When field arithmetic and zero-testing are effective, the matrix, its support graph, and a perfect matching can all be computed in the finite-dimensional degree- n component. $\square$

This is slightly weaker than the especially transparent bijections in Iima–Yoshino Theorem 2.1. Their relations $x_i^p - x_{pi}$ are binomials, so division sends each relevant monomial directly to one monomial and the remainder itself identifies the target partition. For the present ideal, normal forms may contain several standard monomials.

For example, with $\delta = 1 + 2c$, successive reduction in weight 5 gives

$$\text{NF}(x_1^5) = \frac{\delta^3(1+c)^2}{c} x_1 x_4 - \frac{\delta^3(1+c)(2-c)}{c} x_5, \quad \text{NF}(x_1 x_4) = x_1 x_4.$$

Both displayed coefficients are nonzero. Since

$$P(5) = \{(1, 1, 1, 1, 1), (4, 1)\}, \quad Q(5) = \{(5), (4, 1)\},$$

the supported perfect matching necessarily selects

$$(1, 1, 1, 1, 1) \mapsto (5), \quad (4, 1) \mapsto (4, 1).$$

Thus the construction does produce the correspondence claimed after Problem 2.3, but in general the correct procedure is

$$\text{Gröbner division} \quad + \quad \text{finite perfect matching},$$

rather than simply reading a unique monomial from every remainder. Over an effectively presented field, fixing orders on $P(n)$ and $Q(n)$ and choosing the first perfect matching makes this procedure deterministic.

Computational checks

As safeguards against indexing or sign errors, exact computations in $\mathbb{Q}[c]/(c^2 + c - 1)$ verified the syzygy (6) for $2 \leq N \leq 150$. Direct multivariate division verified all 67 non-coprime critical pairs whose least-common-multiple weight is at most 70; every remainder was zero. These finite computations are not used in the proof above.

Novelty status

The Lyness recurrence itself is classical. We have not located a published source connecting it with Iima–Yoshino Problem 2.3 or containing the ideal (5). Bruscek, Mourtada, and Schepers obtain the same monomial ideal in an arc-space setting, but their quotient is the nonreduced arc algebra of a double point rather than a polynomial algebra. Accordingly, the result in this note should be described as apparently new, or “to our knowledge,” until it has received independent expert verification.

References

- [1] C. Bruscek, H. Mourtada, and J. Schepers, *Arc spaces and Rogers–Ramanujan identities*, Ramanujan J. **30** (2013), 9–38. [doi:10.1007/s11139-012-9401-y](https://doi.org/10.1007/s11139-012-9401-y).
- [2] A. N. W. Hone and T. E. Kouloukas, *Deformations of cluster mutations and invariant presymplectic forms*, J. Algebraic Combin. **57** (2023), 763–791. [doi:10.1007/s10801-022-01203-5](https://doi.org/10.1007/s10801-022-01203-5).
- [3] K.-i. Iima and Y. Yoshino, *Gröbner bases for the polynomial ring with infinite variables and their applications*, Comm. Algebra **37** (2009), 3424–3437. [doi:10.1080/00927870802502878](https://doi.org/10.1080/00927870802502878).
- [4] R. C. Lyness, *Note 1581: Cycles*, Math. Gaz. **26** (1942), 62. [doi:10.2307/3606038](https://doi.org/10.2307/3606038).